

17.5 Summary and Notes

17.5.1 Summary

Idea.

1. The cell-means model is the one-way layout on the occupied cells. Any hypothesis $\mathbf{L}\boldsymbol{\mu} = \mathbf{d}$ on occupied cells is tested by Eq. 17.1.1 (Definition 17.1.1, Proposition 17.1.1).
2. Count-weighted marginal means are the most precise, but they estimate the mixture the design happened to produce. Choose the target before the weights (Proposition 17.1.2).
3. An extra sum of squares tests whether the count-weighted residual of the true cell means, after the terms already fitted, is orthogonal to the new term (Lemma 17.2.1). This gives the hypotheses of all the types (Theorem 17.2.2).
4. The last-in sum of squares of a main effect depends on the coding of the *other* factors. It is the Type III sum of squares exactly when they are coded by contrasts (Proposition 17.2.3).
5. Constrained models are cell-means models with $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$, fitted and tested by restricted least squares (Theorem 17.3.1). Under additivity every common weighting of the row means gives the Type II test (Corollary 17.3.2).
6. With empty cells, the interaction has $m - a - b + c$ degrees of freedom, one per independent cycle of the design graph, and the “Type III” sum of squares compares only the complete rows (Theorem 17.4.1).

17.5.2 Notes and sources

Cell means. Analysing unbalanced data through cell means goes back to Yates (1934), whose weighted squares of means test hypotheses about unweighted means. Speed, Hocking and Hackney (1978) identified the cell-mean hypotheses of the sums of squares computed by the programs of their day, and Hocking (1985) and Searle (1987) give book-length treatments. Rencher and Schaalje (2008, chapter 15) presents unconstrained and constrained cell-means models and empty cells, and was the main source for this chapter’s coverage. Christensen (2020, sections 7.5–7.6) treats unbalanced layouts as a problem of comparing models, an emphasis reflected in [Section 17.3](#).

Types of sums of squares. The Type I, II and III terminology comes from statistical software. Nelder (1977) argued from the marginality of model terms that main effects should not be tested in the presence of their interactions, and Nelder and Lane (1995) restated the case. Langsrud (2003) argued for Type II on grounds of power, which Exercise (C1) makes precise. The hypothesis formulas of Theorem 17.2.2(b) and (c) are classical. Christensen (2020, section 7.5) states them for the two-way layout with interaction, and Lemma 17.2.1 is our route to them. The characterization of the coding dependence in Proposition 17.2.3, stated through the orthogonal complement of the other factor’s coding columns, is this book’s formulation of the familiar advice to use sum-to-zero contrasts.

Empty cells. Goodnight (1980) described the construction of the hypotheses behind SAS’s

Type IV sums of squares for designs with empty cells, and noted that they are not unique. Milliken and Johnson (2009) discuss the analysis of two-way layouts with missing cells at length, with an emphasis on estimable questions and on identifying the hypotheses that software tests. The reading of estimable interaction contrasts as the cycle space of the design graph (Theorem 17.4.1(b)) uses the standard fact that a graph with m edges, v vertices and c components has $m - v + c$ independent cycles. The description of the Type III hypothesis with empty cells in Theorem 17.4.1(d) and the software comparison of [Section 17.4](#) are this book's.

Data. The variety trial and the curing study are synthetic, generated by the scripts in `code/ch17/` with fixed seeds, so that the true cell means are known.

17.5.3 References

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